

2022.

Q1.

(a).

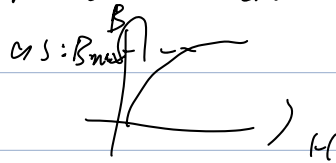
$$\begin{cases} T = k_t \cdot I_a \\ E = k_e \cdot \omega \end{cases}$$

$$\Rightarrow P_{\text{mech}} \omega T = \frac{E}{k_e} \cdot k_t \cdot I_a = \frac{k_t}{k_e} \cdot I_a E = I_a \omega$$

In SI system, $k_e = k_t$

$$\text{then } P_{\text{mech}} = I_a \cdot E = T \cdot \omega$$

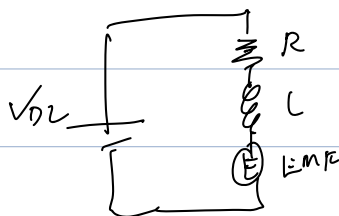
(b) Because the saturation in magnetic materials,



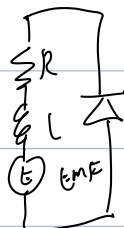
$$V = \frac{d(BS)}{dt} \Rightarrow V \propto f \cdot B \Rightarrow B \propto \frac{V}{f}$$

Q2. (a).

On - state -



off - state



Freewheeling diode is to implement as

where is inductance in the circuit so

in the ON-state, we store energy $\frac{1}{2} L I^2$ in the inductance and release it

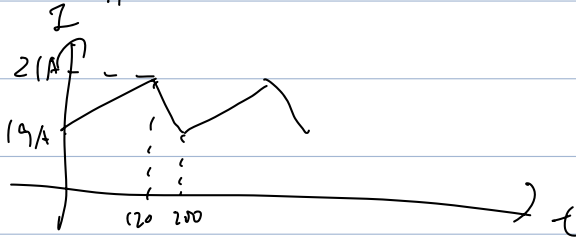
in the off state,

(b).

$$t = \frac{1}{f_{PWM}} = \frac{1}{5k} = 0.2 \text{ ms} = 200 \mu\text{s}$$

$$t_{on} = t \cdot \text{duty cycle} = 0.12 \text{ ms} = 120 \mu\text{s}$$

$$t_{off} = t - t_{on} = 0.08 \text{ ms} = 80 \mu\text{s}$$



$$\Delta I = 21 - 19 = 2 \text{ A}$$

$$I_{AVG} = 20 \text{ A}$$

$$E = k_e \cdot \omega = 1.03 \times 3000 \times \frac{2\pi}{60} = 3\pi \text{ V}$$

ON state: $V_{OL} = L \left(\frac{\Delta I}{t_{on}} \right) + R I_{AVG} + E$

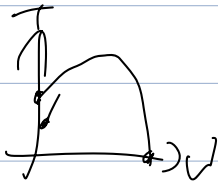
$$\Rightarrow 24 = \frac{2}{120 \mu} L + 20R + 3\pi \quad \textcircled{1} \Rightarrow R = \dots$$

off state: $L \left(\frac{\Delta I}{t_{off}} \right) = R I_{AVG} + E + V_{diode}$

$$\Rightarrow \frac{2}{80 \mu} L = 20R + 3\pi + 0.8 \quad \textcircled{2} \Rightarrow R = \dots$$

$$\Rightarrow \begin{cases} L = 5.952 \times 10^{-4} \text{ H} \\ R = 0.233 \Omega \end{cases}$$

(c). (i).



winding resistor ↓

$$I_{start} = \frac{V}{R_{winding}}$$

↗ $T_{switching}$

(ii).

switching frequency ↑

(iii). magnetic
 $B_{max} \downarrow \Rightarrow R_{ed}$
 airgap

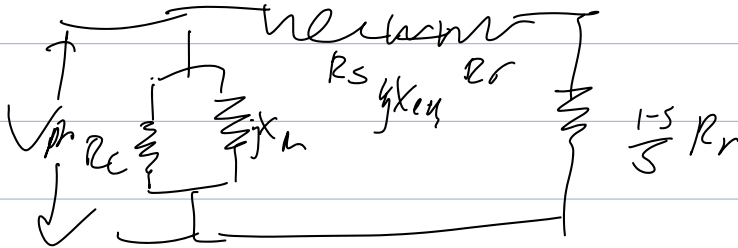
$$\frac{V = E + IR}{L \rightarrow \text{when rotated.}}$$

$E \propto \omega$

Q3.

(a). $P_{mech} = 3 \left(\frac{1-s}{s} \right) R_r \cdot I_r^2$

$$P_{loss} = 3 I_r^2 R_s + 3 I_0^2 R_r + \frac{3 V_{ph}^2}{R_c}$$

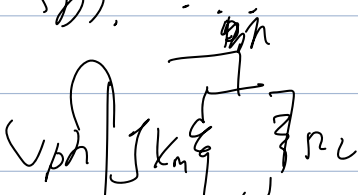


$$\text{Efficiency} = \frac{P_{mech}}{P_{mech} + P_{loss}} \times 100\% = \frac{\left(\frac{1-s}{s} \right) R_r}{R_s + R_r + \frac{1}{R_c} \cdot \frac{V_{ph}^2}{I_r^2} + \frac{1-s}{s} R_r}$$

$$\approx \frac{\frac{1-s}{s}}{1 + 1 + \frac{1-s}{s}} = \frac{1-s}{2s + 1-s}$$

$$= \frac{1-s}{1+s}$$

(b).



1 J

$$(c). N_s = \frac{120 f_s}{P} \Rightarrow f_s = \frac{P N_s}{120} = \frac{6 \times 5000}{120} = 150 \text{ Hz.}$$

$$V_{rms} = \frac{V_{dc}}{2} \cdot \frac{\sqrt{3}}{\sqrt{2}} \cdot M_a \Rightarrow 380 = 0.612 \cdot V_{dc} \cdot M_a$$

$$\Rightarrow M_a = 0.69.$$

$$(d). s = \frac{N_s - f_r}{f_s} = \frac{140}{3000} = \frac{7}{150} \approx 0.0467$$

$$Z_T = R_s + \frac{R_r}{s} + jX_{0a} = 4.523 \angle 7.624^\circ$$

$$|I_{ph}| = \frac{V_{ph}}{Z_T} = \frac{380/\sqrt{3}}{4.523} = 48.506 \text{ A}$$

$$P_{mech} = 3 \times |I_{ph}|^2 \cdot \left(\frac{1-s}{s}\right) R_r = 28.817 \text{ kW}$$

$$= 27.606 \text{ kW} = \omega \cdot T$$

$$\Rightarrow T = \frac{P_{mech}}{\omega} = 96.22 \text{ N.m}$$

$$P_{cu} = 3 I_r^2 (R_s + R_r) + 3 \frac{V_{dc}^2}{R_c} = 2.834 \text{ kW}$$

$$\eta = \frac{P_{mech}}{P_{mech} + P_{cu}} = 91.05\%$$

$$(e). (\because) T_{brake} = -T_{rout} = -96.22 \text{ N.m}$$

$$\text{or } |T_{brake}| = 96.22 \text{ N.m}$$

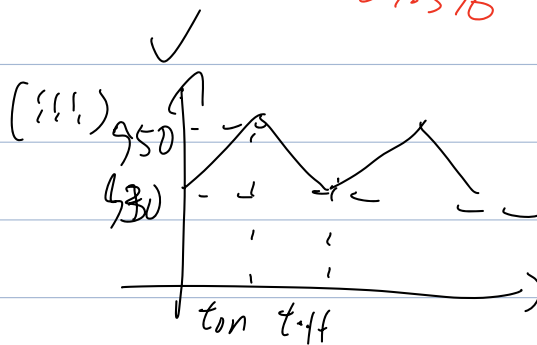
$$(ii). E_{120} = \frac{1}{2} C V_{120}^2 = 16.20 \text{ kJ}$$

$$E_{950} = \frac{1}{2} C V_{950}^2 = 18.05 \text{ kJ} \Rightarrow 2E = 1850 \text{ J}$$

$$P_{break} = 2P_{mech} =$$

$$P_{dc} = (P_{break} - P_{mech} = P_{mech}) = 28.817 \text{ kW} \quad \times 95\% \quad 27.376$$

$$t = \frac{\Delta E}{P_{dc}} = \frac{1850}{27.376} = 67.58 \text{ ms}$$



$$t = \frac{1}{f_s} = 0.15 = 150 \text{ ms}$$

$$\Delta E = \frac{1}{2} k_{comp} (950^2 - 930^2) = 752 \text{ J}$$

$$t_{rise} = \frac{\Delta E}{P_{dc}} = 27.85 \text{ ms}$$

$$t_{fall} = 72.15 \text{ ms} \Rightarrow P_{break} = 34.240 \text{ kW}$$

$$P_{fall} = \frac{\Delta E}{t_{fall}} = 10.423 \text{ kW} = P_{break} - P_{mech}$$

$$= \frac{V_{avg}^2}{R_b} \Rightarrow R_{mech} = \frac{V_{avg}^2}{P_{mech}} = \frac{950^2}{34.240 \text{ kW}}$$

$$= 22.52 \Omega$$

(iv) . . .

Q4.

Calc. $N_s = \frac{120f}{p} = 600 \text{ rpm}$

$$V_{oc} = K_e \cdot \omega = 75.4 \text{ V}$$

= $V_{ph\text{-generator}}$

$$V_{c-r-s} = V_{ph\text{-gen}} \cdot \frac{1}{\sqrt{2}} = 130.8 \text{ V}$$

Then $V_{rectifier} = V_{rectifier} = \sqrt{2} \cdot V_{r.m.s}$
 $= 184.7 \text{ V}$